

HUMAIRA ANZUM NEHA

Lecturer of CSE at UIU || M.Sc. Student at BRAC University || Graduate of KUET

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Faculty Profile Google Scholar LinkedIn GitHub Portfolio

ABOUT ME

Computer Science lecturer and researcher with experience in human-centered AI, computer vision, and accessibility, and strong interests in multimodal AI and natural language processing. My research focuses on designing and evaluating intelligent systems that better support diverse users, particularly those whose needs are often overlooked in conventional technology design.

EDUCATION

2026–Present	M.Sc. in Computer Science and Engineering CGPA: 4.00/4.00; 12 credits completed	BRAC University
2019–2024	B.Sc. in Computer Science and Engineering CGPA: 3.87/4.00	Khulna University of Engineering & Technology (KUET)

RECENT RESEARCH PROJECTS

2026 2nd author	AI-Assisted Speech Editing for People Who Stutter <i>Manuscript under review.</i> - Conducted a qualitative HCI study with eight people who stutter in Bangladesh on video creation and AI-assisted speech editing. - Examined how video context, audience, and personal preferences shaped decisions to preserve or modify stuttering. - Identified the need for AI support that reduces editing effort while preserving creator control and voice representation.
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RESEARCH INTERESTS

Human-Centered AI, Mutlimodal AI, Human-Computer Interaction, Computer Vision, Natural Language Processing, Accessibility

SKILLS

Programming:	Python, C/C++, Java, JavaScript, SQL
AI & ML:	PyTorch, OpenCV, Hugging Face Transformers, Scikit-Learn
Tools & Developments:	Git, GitHub, LaTeX, MySQL, Firebase, Android Studio, ASP.NET

WORK EXPERIENCE

2024–Present	Lecturer in Department of CSE United International University Courses Taught: Database Management Systems, Data Mining, Object Oriented Programming, Computer Architecture, Data Structures and Algorithms.
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RESEARCH EXPERIENCE

2024 1st author	Automated Object Identification for the Visually Impaired  <i>Published in the 2024 27th International Conference on Computer and Information Technology (ICCIT).</i> - Developed an edge-compatible assistive system for common-object detection, Bangladeshi currency and denomination recognition, and face recognition. - Collected and annotated custom image datasets for model training and evaluation: object-detection dataset and Bangladeshi currency dataset. - Supervisor: Dr. Md. Sheikh Sadi, Professor of CSE, KUET. - Technologies Used: Python, PyTorch, OpenCV, YOLOv5.  Repository
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2024	An Assistive Device for the Visually Impaired for Enhanced Navigation and Reading  <i>Published in the 2024 13th International Conference on Electrical and Computer Engineering (ICECE).</i> - Developed embedded spectacle glasses with real-time obstacle avoidance, fall detection, and textual information extraction modules. - Supervisor: Dr. Md. Sheikh Sadi, Professor of CSE, KUET. - Technologies Used: Python, PyTorch, OpenCV, YOLOv5, Raspberry Pi, OCR, Accelerometer, and ToF Distance Sensor.
2023 1st author	Mobile Application to Collect Data and Measure Blood Component Levels in a Non-invasive Way  <i>Published in the 2023 26th International Conference on Computer and Information Technology (ICCIT).</i> - Developed a mobile application for non-invasive measurement of blood components. - The user places their fingertip on an NIR LED device, and a 15-second video is recorded. The video is analyzed to extract PPG signals, which are processed through a DNN model to predict glucose and hemoglobin levels. - Supervisor: S M Taslim Uddin Raju, Lecturer of CSE, KUET (on leave). - Technologies Used: Kotlin, Android Studio, Python, Photoplethysmography (PPG) Signal Analysis, and OpenCV.

OPEN DATASETS AND RESEARCH RESOURCES

2026	MusicCaps-Curated-Tags  Curated MusicCaps annotations into 216 standardized labels across genre, instrument, mood, vocal, and tempo categories using an LLM-assisted pipeline with manual review and validation.
2024	Computer Vision Datasets Assistive Object Detection Dataset  (approximately 21,000 YOLO-annotated images across eight classes) and Bangladeshi Currency Classification Dataset .

ACADEMIC PROJECTS

Python, Pygame	Abalone AI  Developed a playable Abalone board game in Pygame with an AI opponent powered by Minimax and alpha-beta pruning.
Bootstrap, JavaScript, ASP.NET	Smart&Safe  Developed a web app for buying and selling used smartphones with identity verification (NID + security information) and automated IMEI screening through an admin dashboard.
Java, Firebase, Android Studio	Together  Created a task and deadline management mobile application with collaboration and communication features for teams and coworkers.

ACHIEVEMENTS

3 times	Dean's List Award  Awarded for consistent academic excellence and strong semester-wise academic performance.	Khulna University of Engineering & Technology (KUET)
2021	Top-10 Finalist  Designed and prototyped a community-based application idea selected among the hackathon's top 10 entries.	UNDP Women's Digital Innovation Hackathon 2021
2018–2019 to 2021–2022	Top-3 CGPA (among 120 students) Ranked in the top 3 by cumulative academic performance within the cohort.	Khulna University of Engineering & Technology (KUET)